

Reconstructing a language with gaps

A method of controlled construction, its measurements, and its application to Merovingian Franconian

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Abstract. Reconstructing a language whose witnesses are full of gaps poses a problem of method before it poses one of linguistics: at every empty slot one must choose between leaving a hole and inventing — and nothing, in the result, afterwards distinguishes what was attested from what was filled in. This document describes an apparatus that makes that distinction *verifiable*, and applies it to Merovingian Franconian, the language of the court of Clovis, of which neither text nor grammar survives. The cycle runs as follows: **source registers** generate the whole documentation; **40 automatic controls** check its consistency, each declaring what it does not see; **blind translations** — two independent readers, the same files, no contact — measure whether the documentation suffices to produce a text without inventing anything; and what they reveal goes back into the registers, or stays **declared open**. The state so reached counts **1751 lemmas** and **353 verbs**; over the last 18 rounds the translators produced **2 invented form** in all, for an average of 9.0 slots where the documentation is silent. The second part describes the language itself — writing, sound laws, inflection, syntax — and two hard decisions are followed there from the raw witness to the form adopted.

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Part One — The project

1. The problem of method

A language whose witnesses are full of gaps is not reconstructed by stopping where they stop: a target language for translation must be able to say what is asked of it. One must therefore fill in — and the danger is not the filling in, it is that **nothing afterwards distinguishes the filling from the attestation**. A system invented from end to end can be perfectly consistent, and its very consistency makes it credible.

The apparatus described here does not claim to avoid filling in. It makes it *traceable*: every form carries its status, every ruling its date and its witness, and what could not be settled stays written as unsettled. The cycle turns in five stages.

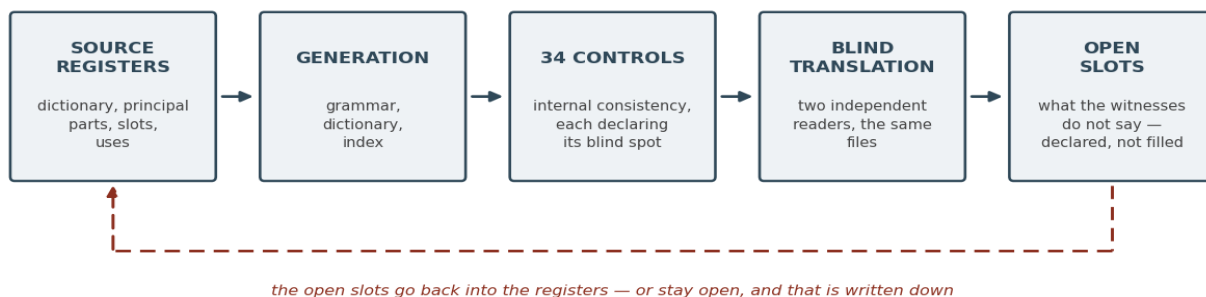


Figure 1 — The cycle of construction. Nothing in what is delivered is written by hand: the registers are the only source, and the documentation is generated from them. The blind translations are the only point at which the apparatus is set before a reader who does not know its intentions.

stage	what it does	what it does not do
source registers	four files hold everything: dictionary, principal parts, inflectional slots, uses	they are never read directly by a translator
generation	the delivered grammar, dictionary and index are derived from them at every pass	no correction is ever made in the generated file
controls	check internal consistency : abrogated forms that survive, slots declared empty but filled elsewhere, lemmas that have become unreachable	⚠ they check nothing of historical truth , and each declares its blind spot
blind translation	measures whether the documentation suffices — two separated readers: do they converge? did they invent?	does not measure whether the language is right: a false but well-documented system would give the same curves
open slots	what the translators had to guess goes back into the registers, or is entered there as unsettled	one does not fill a slot in order to make a red go away

⚠ **It is the fourth stage that carries the whole probative weight.** The first three are documentary engineering: they guarantee that a system does not contradict itself. Only the blind translation brings in a judge who did not take part in the construction — and it is the one measurement the project cannot rig without noticing.

2. The object — Franconian

Franconian is the Germanic language of the Salian and Ripuarian Franks, spoken between the Scheldt and the middle Rhine before it gave way to Romance. It is known only obliquely. No continuous text survives; what is known of it comes from four indirect sources: the latinised personal names of the diplomas and the chronicles, the so-called *Malberg* glosses of the *Lex Salica*, the loans into Gallo-Romance — *guerre*, *garder*, *heaume*, *franc* — and comparison with its better preserved neighbours.

The project does not claim to restore that language. It builds a **usable reconstruction** of it: a coherent, dated, documented system in which an outside reader can translate without inventing forms. The distinction is kept everywhere, and it is the first thing the project writes about itself.

3. The witnesses

Four corpora, none satisfactory on its own.

rank	witness	what it is
(a)	Old Low Franconian — the Wachtendonck psalters	the same area, the same filiation — but three centuries later, and an interlinear translation of a Latin original
(a bis)	Merovingian Latin — the glosses of the <i>Lex Salica</i> , personal names, the Bergakker inscription	contemporary with the target , but filtered through a Latin orthography and a Latin setting
(b)	Old Saxon — the <i>Heliand</i>	continuous Germanic syntax — but coastal, three centuries later, and in alliterative verse
(c)	Old High German	richly described and dated — but it underwent the second consonant shift that the target refuses
(d)	Gothic	an archaic check, outside the area

⚠ **Old English is excluded as a support.** It shares enough features with Franconian for one to draw on it without noticing, and it diverges from it on the very points that decide.

The rank depends on the domain

A single global rank would suggest that one witness prevails in the same way on every question. **That is false:** a corpus can be excellent for the lexicon and worthless for word order. The ranking is therefore established by domain.

domain	which prevails	why
phonology	(a bis), then (a)	Merovingian Latin is contemporary ; Wachtendonck is three centuries later — but see the reservation below
spelling	(a)	a continuous Germanic text, written to be read as Germanic
morphology	(a), then (b)	Wachtendonck really declines : the only witness of the area that gives paradigms in context
syntax	(b), then (c)	△ rank (a) is the worst here : an interlinear gloss whose word order copies the Latin
lexicon	(a bis), then (a)	the Malberg glosses name real things, with no Latin original to trace

△△ **Practical consequence:** on a question of word order, Wachtendonck is no authority. Its government and its order are those of its Latin source.

Chronological nearness is not phonographic fidelity

The project long held that Latin scribes “have no reason to normalise a Germanism”. **That is too strong, and the reservation is asymmetrical.**

vowel	is the Latin testimony worth anything?
ai	yes. Latin no longer had a living digraph (ai) — (ae) had been worth /ɛ/ since the 2nd century. An (ai) under a Merovingian pen transcribes a foreign sound : <i>Gais-</i> , <i>Rait-</i> , <i>Bainobaudes</i>
au	no. Latin had (au) and was monophthongising it in Gallo-Romance — the <i>Appendix Probi</i> records the confusion (<i>auris non oricla</i>). A Merovingian (au) may be a real /au/ or a graphic hypercorrection for /o/

△ Rank (a bis) therefore has chronological nearness without reservation; it has phonographic fidelity only where Latin has no sound of its own to put in the place. *This reservation overturns no ruling already given — that on the diphthongs rests on the Riparian descent and on Braune’s dating, not on the personal names alone — but it changes what may be pleaded.*

Two rules govern the use of these tables. **Two witnesses that diverge are never averaged:** the divergence is declared and the slot stays open. And **internal consistency is not a dating** — the project learnt that on eleven preterites which confirmed one another and which were eleven times the same mistake.

4. The registers and the controls

The delivered documentation is **generated**, never kept by hand. Four source registers — the dictionary, the principal parts, the inflectional slots, the uses — feed a chain that produces the grammar, the dictionary and the index.

40 automatic controls run at every pass. They do not check the language — no program can — but its *internal consistency*: that an abrogated form does not survive elsewhere, that a slot declared empty is not filled further on, that a lemma reachable yesterday still is, that a strong class does not carry two ablaut vowels.

△△ **One of them reads what a TRANSLATOR writes, and it is the only one.** The other thirty-five weigh the files the project publishes — they measure the apparatus against itself. The translation is the one place where the documentation is put to work, and until the 80th round nothing looked at it. The copies handed in are now kept and re-judged at every generation: a rewording of one entry can withdraw the warrant of a form ten translators wrote, and the kit then stops supporting a text it used to support — silently, since the copies are already written and it is the NEXT round that pays.

△ **Every control declares what it does not see**, and those declarations are gathered on a single page. It is the only honest guarantee such an apparatus can offer: it measures consistency, and consistency is not truth.

5. The protocol of validation

The double enquiry. When a question cannot be settled internally, two enquirers receive the same file, work separately, and are not allowed to open the project's own files — they enquire into the attested languages, not into what a project has written about them. They are asked to rate their answer, to name what they could not establish, and to write “reference not verified” rather than invent a citation.

The blind round of translation. Two translators receive a fresh text and the three delivered files, nothing else. What is measured is not the beauty of the rendering but the *convergence* — do two readers with no contact produce the same text? — and the number of forms they had to invent.

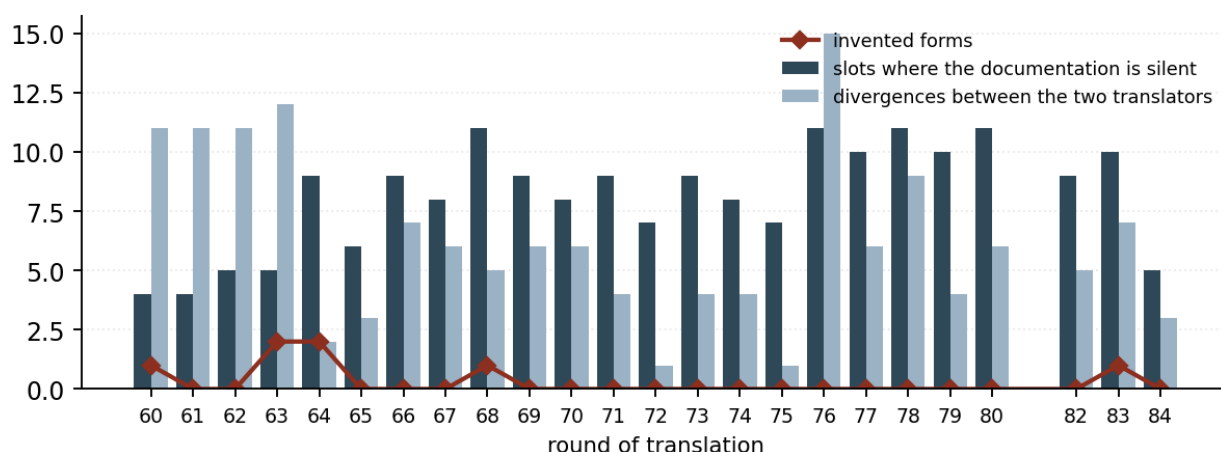
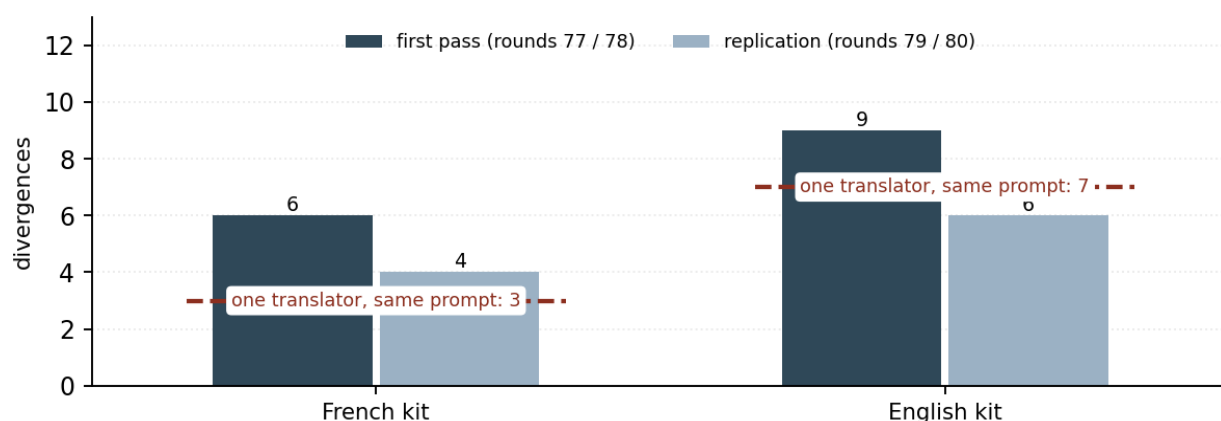


Figure 2 — The last 24 rounds. The bars count the slots where the documentation is silent and the divergences between translators; the line counts the invented forms. The blockages do not decrease: every round explores a fresh register, and a fresh text always finds silences. The inventions, for their part, have stopped.



The documentation was translated, and the comparison was controlled. The whole apparatus was written in French and turned to English; the question was whether the delivered files still hand on as much. One fresh text was written in BOTH languages with the same content, and run on the last wholly French state of the kit and on the English one, two translators each. **Zero invented forms in the four copies**, the same walls met on both sides — no word for “chain”, no noun of silence, the same verb silent on its second argument — and the same two declared collisions found independently. The kit came through the translation.

And the comparison was then replayed, which is the part worth printing. Same text, same brief, fresh sessions: the counts fell from 6 and 9 to 4 and 6, while a single translator moved in **seven** places between two runs of the same prompt. The spread inside a condition is the size of the spread between conditions, so the gap between the two kits measures nothing. **What reproduces is not the count but the place:** round after round the divergences fall on the slots the documentation itself declares open. That is what a divergence bar reads — a map of the silences, not a score.

Figure 3 — The controlled comparison and its replication. The dashed line is the distance a single translator travelled between two runs of the SAME prompt; it stands above the gap the bars are about. A measurement whose noise exceeds its signal is a reading, and must be printed as one.

6. The measured state

measure	value
lemmas in the dictionary	1751
entries in the index	4692
verbs	353
... of which the four principal parts are written	336
... of which the government is recorded	286
nouns with an inflectional class assigned	349
coverage of the vocabulary of the everyday scene	196 / 196
automatic controls	40
open slots declared	100

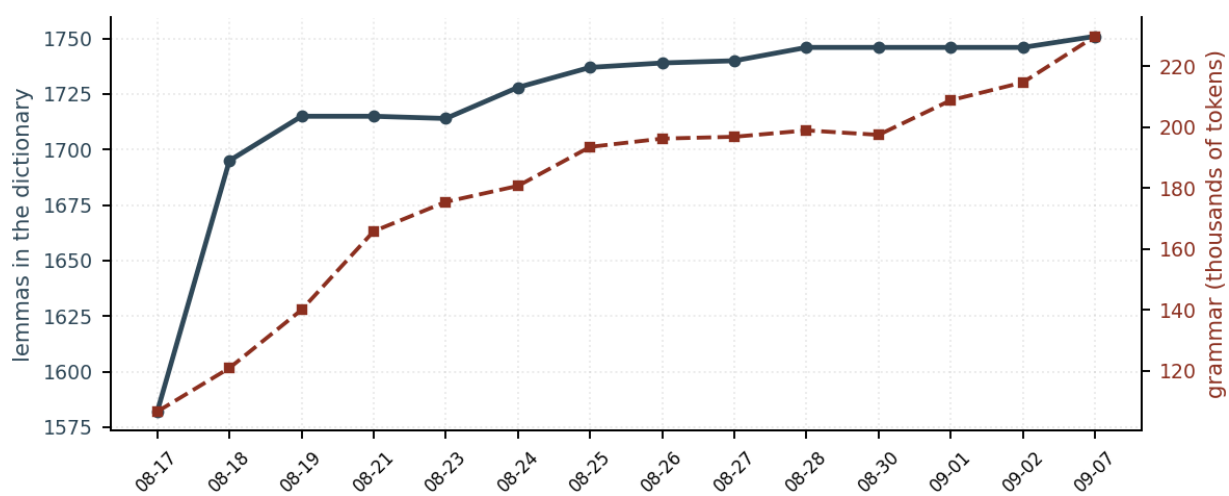


Figure 3 — Growth of the dictionary and of the grammar. The grammar grows much faster than the lexicon: it accumulates not words but adjudications, each dated and motivated. That is the cost of the probative discipline, and it is visible.

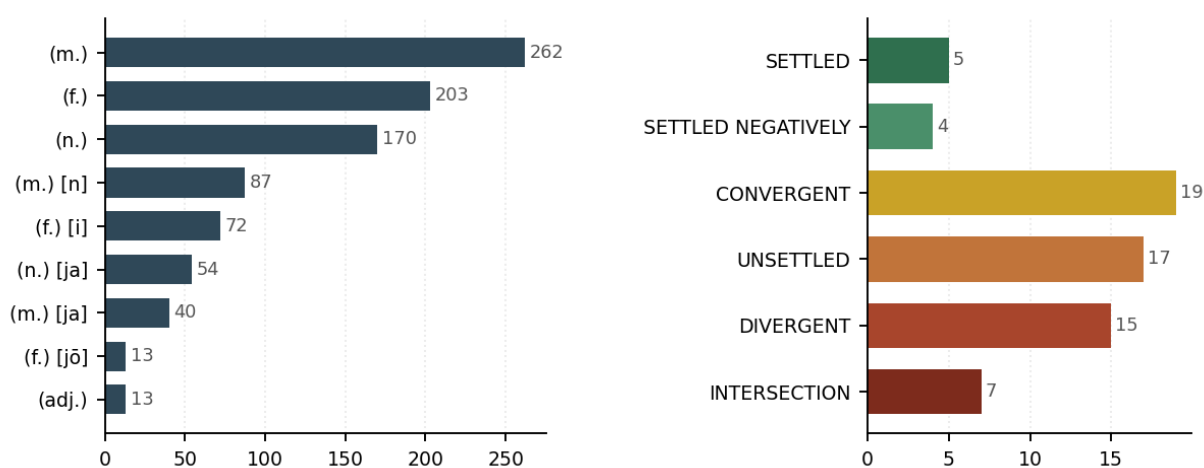


Figure 4 — On the left, the inflectional classes of the noun by size. On the right, the state of the recorded uses: what two translators rendered in the same way, differently, or not at all.

7. Result

Over the last 18 rounds — six registers: past narrative, sentence in the present, injunction, reported speech, irrealis, description — the two independent translators produced **2 invented form in all**, and rendered texts often identical word for word over several consecutive sentences. The best round gave seven identical sentences out of eight.

That is the only result this project claims, and it is a modest one: **the language is usable**. It does not follow that it is Franconian.

8. The status of each major rule

A rule of the project can have three statuses, and the distinction matters more than the rule itself. **Attested**: a witness carries the form. **Inferred**: it is deduced by a dated argument from witnesses that do not carry it. **Conventional**: the witnesses do not settle it, and the project chose, and declared that it was choosing.

rule	status	support
no second shift — <i>tehan, skip</i>	attested	the very definition of the area; rank (a) writes it everywhere
loss of final -z (L1)	attested	all the western witnesses
accusative = nominative in the -a stems	attested	fifteen examples from the internal corpus, the article alone carrying the case
V2 in main clauses, verb final in subordinate clauses	attested	the <i>Heliand</i> and the <i>Lex Salica</i> ; Behaghel
3rd sg. of the copula = <i>ist</i>	inferred — 70 to 80 %	Wachtendonck 36:0 and 38:0 across two editions; the direction of change (the -t is inherited)
the diphthongs <i>ai</i> and <i>au</i> intact	inferred — 85 to 88 %	Braune §43 Anm. 1 and §45 Anm. 2; the Wissembourg charters (<i>Gairoaldo</i> , 700); the Ripuarian descent
clitics in second position	inferred — measured	Wackernagel's law; an internal measurement on the <i>Heliand</i> — 9.4 % after <i>endi</i> against 1.9 % after <i>thō</i>
the weak adjective = individualisation, not “after the article”	inferred	the determiner is not obligatory at 550: the modern German rule is logically impossible
the preterite plural distributed -um / -ut / -un	inferred	rank (a), against the syncretism of the <i>Heliand</i>
the spelling (eu) rather than (eo)	conventional — 52 %	a double enquiry declared undecidable ; the project keeps (eu) and no longer claims to prove it
1st and 2nd pl. of the copula	conventional	zero attestations; three possible systems, no form common to two ranks

△ **An outside reader must be able to separate “a rule dated within the project” from “a philological argument verifiable outside it”.** The lines marked *attested* and *inferred* point to a witness or to a paragraph of a grammar; the *conventional* lines point only to a decision, and say so.

9. What the witnesses do not say

100 slots remain open, declared as such. They are not all debts: several of them are frontiers.

open slot	why it stays open
the date of the coastal monophthongisation	no <i>terminus post quem</i> — the witnesses do not give it

open slot	why it stays open
the infinitive of class II — <i>beugan</i> / <i>biogan</i> / <i>biugan</i>	a decision about the area, rated 50 % by the enquiry
the first two persons of the plural of the copula	zero attestations at rank (a); three possible systems, no form common to two ranks
the mood of the indirect question	two translators, two arguments equally founded on an internal rule of the kit
the spelling (eu) against (eo)	a double enquiry declared undecidable at 52 %: the kit keeps (eu) as a convention and no longer claims to prove it

10. Limits

This is not a scholarly reconstruction. A scholarly reconstruction stops where the witnesses stop; this one goes on, because a target language for translation must be able to say what is asked of it. It therefore fills in — but under constraint, dating every filling and distinguishing everywhere the attested, the derived and the agreed.

Internal consistency is not proof. A false system can be perfectly consistent: the project verified that at its own expense on a series of eleven forms that confirmed one another.

The rounds do not measure historical truth. They measure whether the documentation suffices to produce a text without inventing. A wholly false but well-documented Frenkisk would give the same curves.

Part Two — The language

What follows is a description for use: enough to read and write Frenkisk. The paradigms and the tables are extracted from the project's registers at every generation of this document; the explanations are written.

11. Writing and pronunciation

sign	value	example
ā ē ī ō ū	long vowels — the macron is distinctive, it is no ornament	dāl “share” ≠ dal “valley”
th	the dental spirant <i>p</i> , as in English <i>thing</i> ; the project writes it th	theud “people”
hw	an initial aspirate, as in English <i>wh-</i>	hwē “who?”
k / qu	<i>k</i> everywhere; qu before the raised vowel	kuman but quimid
ai au eu iu	diphthongs — ai and au are full at 550, they have not yet been reduced	stain , baum , beudan

△ **There is no second consonant shift.** The stops *p*, *t*, *k* stay intact: one writes **tehan** “ten” and **skip** “ship”, never the High German forms *zehan*, *skif*. It is the feature that most visibly separates Frenkisk from German.

12. The sound laws

Ten numbered laws govern the passage from Common Germanic to Frenkisk. **Two have been abrogated** by enquiry: they described a state later than the target.

law	what it does	example
L1	loss of final <i>-z</i>	dagaz → dag
L2	abrogée	stainaz → stain · gaitiz → gait
L3	abrogée	baumaz → baum
L4	<i>eu</i> → <i>iu</i> before <i>i</i> , <i>j</i> , <i>u</i> ; <i>eu</i> everywhere else	beudan → biudis ~ beudan
L5	umlaut <i>a</i> → <i>e</i> before <i>i</i> / <i>j</i>	gastiz → gast, pl. gesti
L6	<i>p</i> kept, written th	þeudō → theud
L7	NO second shift: <i>p t k</i> intact	tehan, skip
L8	short final vowels kept	sunu, gebu
L9	<i>ē</i> → <i>ā</i>	māno
L10	final hardening of the labial spirant: <i>*b</i> → <i>f</i> in absolute final position, <i>b</i> before a vowel...	wīf / wības

△ **The abrogation of L2 and L3 is the project's chief ruling.** The kit long wrote *stēn*, *bōm*, with long vowels drawn from diphthongs. A double enquiry established that the monophthongisation is a hundred and fifty to two hundred years later than the target: Braune dates the contraction of *ai* to the 7th century and that of *au* to the years 750-760, and the Wissembourg charters still carry *Gairoaldo* in 700. One therefore writes **stain**, **baum**, and **staig** in the preterite.

13. The noun

The noun declines in four cases — nominative, accusative, genitive, dative — in the singular and the plural, plus a residual instrumental. **349 nouns** carry an assigned class; here are the four main ones.

a-masc dag “day”		a-neuter word “word”		ō-fem geba “gift”		n-weak Franko “a Frank”	
slot	form	slot	form	slot	form	slot	form
nom. sg.	dag	nom. sg.	word	nom. sg.	gebu	nom. sg. masc.	Franko
acc. sg.	dag (internal)	acc. sg.	word (internal)	acc. sg.	geba		
gen. sg.	dagas	gen. sg.	wordas	gen. sg.	geba	nom. sg. neuter	herta
dat. sg.	dagē	dat. sg.	wordē	dat. sg.	gebu	obliques sg.	-on
instr. sg.	dagū	nom. pl.	word	nom. pl.	gebā	nom./acc. pl.	-on
nom. pl.	dagā	acc. pl.	word	acc. pl.	gebā	gen. pl.	-ono
acc. pl.	dagā (internal)	gen. pl.	wordō	gen. pl.	gebōnō ~ gebō	dat. pl.	-on
gen. pl.	dagō	dat. pl.	wordum	dat. pl.	gebōm		
dat. pl.	dagum						

△ **The accusative singular is not distinct from the nominative** in the *-a* stems: it is the article that carries the case, *the dag / thena dag*. The *-ō* feminine, however, does distinguish them — and that is a trap: **the dictionary cites those words in the accusative** (*geba*), their nominative is in *-u* (*gebu*). To copy the lemma for a subject is to write an accusative.

14. The adjective — the least intuitive rule

series	when to use it	example
weak (-o, -on)	when the adjective individualises — “that one, as against the others”	gōdo man “the good man”
strong	everywhere else, including in indefinite reference	gōd man “a good man”
predicate	always strong ; and the comparative is bare there, uninflected, whatever the gender and number	forhtu ist hardiro “fear is harder”

△ **It is not “weak after the article”**. That rule is the modern German one, and it is logically impossible at 550: the determiner is not obligatory there. **It is the weak adjective that carries the individuation**; the article only adds a deixis, and it remains optional. *Aldo hund* is enough to say “the old dog”.

15. The verb

The strong verb marks tense by **ablaut**: the vowel of the root changes. Four forms suffice to conjugate it — infinitive, preterite singular, preterite plural, participle. The register records **353** of them, of which **336** with their four principal parts written.

class	verbs	infinitive	pret. sg.	pret. pl.	participle
I	25	bīdan	baid	bidum	gabidan
II	19	beudan	baud	budum	gabodan
III	33	bergan	barg	burgum	gaborgan
IV	13	beran	bar	bārum	gaboran
V	24	geban	gaf	gābum	gageban
VI	25	bakan	bōk	bōkum	gabakan
VII	27	bannan	bēn	bēnum	gabannan

△ The present raises its vowel throughout the singular, never in the plural. *geban* makes **gibu** / **gibis** / **gibid** but **gebam** / **gebid** / **geband**; *beudan* makes **biudu** / **biudis** / **biudid** against **beudam**. It is law L4 for class II (*eu* → *iu*) and the raising *e* → *i* for classes III to V.

The weak verbs

Two living classes; the preterite there is made by a dental suffix, not by ablaut.

class	infinitive	preterite	participle	sense
weak I (-ian)	haurian	haurda	gahaurid	“to hear”
weak II (-ōn)	wardōn	wardōda	gawardōd	“to watch”

△ The **weak class III** in *-ēn* has been **dissolved** by ruling: it no longer exists, and one does not write *†natēn* “to become wet”. **There is no inchoative suffix at all**: “to become X” is said **werthan** + the adjective — *wirthid nat*, *werthand alda* “they grow old”.

What the verb does not have

No future: the present carries it. **No imperfect**: a single preterite, which renders “he was carrying” as “he carried”. **No conditional**: the preterite subjunctive serves for the irrealis, and it does not distinguish “I would not come” from “I would not have come”. **No morphological passive**: *werthan* + participle for the process — **wirthid gabrokan** “is being broken” — and *wesan* + participle for the state, **ist gabrokan** “is broken”.

16. The syntax

rule	what it says
V2 in main clauses	the finite verb takes second place, whatever constituent opens: <i>Thanne gildis thū</i> “then you will pay”, where the subject moves behind the verb
verb final in subordinate clauses	and the bound holds strictly in the legal register: <i>of ain man sīn rōthar lātīd</i> “if a man leaves his oar”
negative V1	a clause opened by <i>nī</i> puts the verb first: <i>Nī sprak hē mair</i> “he spoke no more”
clitics in second position	the unstressed pronouns group after the first stressed constituent, and the group is not broken up
endi is in position zero	the coordinator does not count in the reckoning: what follows it is a whole clause

△ **There is no “A, B and C” pattern**. The final coordinator is a modern habit; a series of three elements or more is written by **asyndeton** — *thrī ohson, twē swīn, nīgun skāp*. *endi* is reserved for two phrases, or for two verbs under a single subject.

△ **The article is never required by the syntax.** Frenkisk puts in about one where a modern language puts three or four: generic, mass and abstract nouns are bare, and a legal enumeration is bare from end to end.

17. Three texts analysed

Each of them was translated by two independent readers who had only the three delivered files. What follows is what the two agreed upon.

A sentence in the gnomic register

Smith sīn īsarn kiusid, raginburgio sīna word kiusid.

The smith chooses his iron, the judge chooses his words.

form	analysis
kiusid	<i>keusan</i> (class II), 3rd sg. of the present — the vowel raised by L4: <i>eu</i> → <i>iu</i>
sīn, sīna	the reflexive possessive: the possessor is the subject. If it were somebody else, <i>is</i> would be needed, invariable — and the sentence would stay impeccable while naming a different person
smith, raginburgio	bare, in generic use: a modern language puts two articles, Frenkisk none
the order	the verb final in both members — that is the gnomic register, not subordination

An irrealis

Of ik wissi, ni quāmi ik. Aldo hund wāri noh kwik.

Had I known, I would not have come. The old dog would still be alive.

form	analysis
wissi, quāmi, wāri	preterite subjunctive, in the protasis and in the apodosis. One single tense carries the irrealis of the present and that of the past
Of + subjunctive	an exception to the rule “ <i>of</i> in the protasis keeps the indicative”, which aims at the real protasis: the indicative would give as real what a counterfactual declares false
Aldo hund	a weak adjective with no article — it is enough to say “the old dog”. And it is the adjective that carries the gender: <i>hund</i> alone also means “a hundred”
ni quāmi ik	negative V1: <i>ni</i> opens, the verb follows, the subject closes

A description

Kald fiur, īdal beud, thrī enga bankā.

A cold fire, an empty table, three narrow benches.

form	analysis
the asyndeton	no <i>endi</i> : it is a series of three, and the final coordinator does not exist in Frenkisk
kald, īdal, enga	attributives in the strong series, the reference being indefinite; <i>enga</i> loses its <i>-i-</i> before the plural ending
thrī	the numerals 1 to 3 decline; from 4 on they are invariable and the noun keeps its case
the absence of a verb	the nominal sentence is the style of enumeration; nothing obliges one to add a copula

18. Two hard decisions, from the raw witness to the form

What the tables do not show is the road. Here are two rulings followed step by step.

Case no. 1 — the preterite of “to climb”: *stēg* or *staig*?

stage	what happened
the initial state	the dictionary carried <i>stēg</i> , with three other monophthongised forms: <i>bōg</i> , <i>frōs</i> , <i>slōt</i>
the report	an internal control had been printing them as “out of series” at every run for ten days — reported and tolerated
the internal counter-argument	one might have believed in a conditioning: three of them fall before a dental, the classic environment of monophthongisation
what refuted it	△ each form had a twin in the same register — same class, same following consonant — that carried the diphthong: <i>kaus</i> against <i>frōs</i> , both before <i>s</i> . No conditioning produces that: the condition is identical
the witnesses	a double enquiry. Braune §43 Anm. 1: the Wissembourg charters carry <i>Gairelaigo</i> in 696 and <i>Gairoaldo</i> in 700 — the diphthong still holds before <i>r</i> , the environment where it falls earliest. §45 Anm. 2: <i>au</i> does not pass to <i>ō</i> until 750-760
the decisive argument	the target is defined as the common ancestor of Dutch and of Ripuarian. Now Ripuarian continues a <i>conditioned</i> treatment, which requires a diphthong as input : no mechanism re-diphthongises an <i>ē</i>
the check	three of the four forms were attested nowhere ; the fourth was attested in Old Saxon alone — the language the charter forbids following to the letter
the ruling	staig, baug, fraus, slaut . Rated 85 % and 88 %

Case no. 2 — the copula in the 3rd person: *is* or *ist*?

stage	what happened
the initial state	the kit wrote <i>is</i> in all its worked examples, and <i>nīs</i> as the contraction
the survey	Wachtendonck, exhaustively, in two editions: ist 36 times (Cowan 1957) and 38 times (Heyne 1867). <i>is</i> : none . The only <i>is</i> known is a conjecture of van Helten in a lacuna
the argument from direction	the <i>-t</i> is inherited — *h ₁ és-ti, Lat. <i>est</i> , Goth. <i>ist</i> . It is the form WITHOUT <i>-t</i> that is the innovation, and its distribution is coastal: OS <i>is</i> , OFris <i>is</i> , ON <i>es</i>
the finest check	one enquirer separated the two layers of the psalter: 37 of the 38 <i>ist</i> are in the Low Franconian layer , not in the Middle Franconian model. It is therefore no residue of the source: it is the form of the transposer himself
what the ruling cost	the kit’s internal consistency argued for <i>is</i> — but it argued only for Old Saxon , on which the kit had aligned itself without knowing. <i>A kit consistent with the wrong witness is consistent</i> .
what stays open	the contraction <i>nīst</i> is derived, never attested at rank (a); and the western fringe is not covered — rated 80 % in the east, 58 % in the west
the ruling	ist . Rated 70 % and 80 %

Appendix — the coverage test set

The measure “coverage of the everyday scene” means nothing unless its protocol is known.

property	state
nature	△ it is not a frequency list . A word frequent in a modern language measures the distance to that language; a word called for by a scene measures what the language can do
constitution	words required by ten narrative scenes chosen for what the <i>Pactus</i> does not cover: a quarrel between neighbours, a meal, a sick child, a market, a welcome, a suit carried to verdict, a river crossing, a long illness, a cattle market, a vigil
closure	the list was closed on 17 August; extended once only , on 18 August, from 131 to 196 words — and the extension is journalled with its motive

property	state
motive for the extension	the list showed 131/131 = 100 % . A saturated list measures nothing any more: it only certifies that one has finished going down the list one had written
stability since	the denominator has been frozen at 196 since 18 August. $\triangle$ And the numerator has fallen in the interval — 196, then 193, then 194: the list is not adjusted to flatter the result
where it is	in tools/mesura.py , in plain sight, with its journal of extension

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